

Posterior Probability in Gaussian Mixture Models

The probability that $x^{(i)}$ is being modeled from the j -th cluster out of the k clusters is given by the posterior probability:

$$p(z^{(i)} = j \mid x^{(i)})$$

Using Bayes' Rule, this is computed as:

$$p(z^{(i)} = j \mid x^{(i)}) = \frac{p(x^{(i)} \mid z^{(i)} = j) \cdot p(z^{(i)} = j)}{\sum_{\ell=1}^k p(x^{(i)} \mid z^{(i)} = \ell) \cdot p(z^{(i)} = \ell)}$$

In a Gaussian Mixture Model:

- $p(z^{(i)} = j) = \phi_j$ is the prior probability of choosing cluster j ,
- $p(x^{(i)} \mid z^{(i)} = j) = \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j)$, the Gaussian likelihood under cluster j .

Substituting, we get:

$$p(z^{(i)} = j \mid x^{(i)}) = \frac{\phi_j \cdot \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j)}{\sum_{\ell=1}^k \phi_{\ell} \cdot \mathcal{N}(x^{(i)}; \mu_{\ell}, \Sigma_{\ell})}$$

This is called the **responsibility** of cluster j for data point $x^{(i)}$, and is typically denoted as:

$$\gamma_j^{(i)} := p(z^{(i)} = j \mid x^{(i)})$$

Understanding ϕ_j

The parameter ϕ_j is the prior probability that a randomly chosen data point comes from the j -th cluster:

$$\phi_j = p(z^{(i)} = j), \quad \text{with} \quad \phi_j \geq 0, \quad \sum_{j=1}^k \phi_j = 1$$

In the EM algorithm, ϕ_j is updated in the M-step as:

$$\phi_j = \frac{1}{m} \sum_{i=1}^m \gamma_j^{(i)}$$

where m is the total number of data points and $\gamma_j^{(i)}$ is the responsibility of cluster j for data point $x^{(i)}$.

We wish to model the data by specifying a joint distribution $p(x^{(i)}, z^{(i)}) = p(x^{(i)}|z^{(i)})p(z^{(i)})$. Here, $z^{(i)} \sim \text{Multinomial}(\phi)$ (where $\phi_j \geq 0$, $\sum_{j=1}^k \phi_j = 1$, and the parameter ϕ_j gives $p(z^{(i)} = j)$), and $x^{(i)}|z^{(i)} = j \sim \mathcal{N}(\mu_j, \Sigma_j)$. We let k denote the number of values that the $z^{(i)}$'s can take on. Thus, our model posits that each $x^{(i)}$ was generated by randomly choosing $z^{(i)}$ from $\{1, \dots, k\}$, and then $x^{(i)}$ was drawn from one of k Gaussians depending on $z^{(i)}$. This is called the **mixture of Gaussians** model. Also, note that the $z^{(i)}$'s are **latent** random variables, meaning that they're hidden/unobserved. This is what will make our estimation problem difficult.

The parameters of our model are thus ϕ , μ and Σ . To estimate them, we can write down the likelihood of our data:

Explanation

We aim to model the data by introducing a joint distribution over both the observed data point $x^{(i)}$ and an unobserved (latent) variable $z^{(i)}$. Specifically, we factor the joint distribution as:

$$p(x^{(i)}, z^{(i)}) = p(x^{(i)} | z^{(i)}) \cdot p(z^{(i)}).$$

Latent Variable Model

We define:

$$z^{(i)} \sim \text{Multinomial}(\phi),$$

where $\phi = (\phi_1, \phi_2, \dots, \phi_k)$ is a probability vector such that:

$$\phi_j \geq 0 \quad \text{and} \quad \sum_{j=1}^k \phi_j = 1.$$

This means that $z^{(i)} \in \{1, 2, \dots, k\}$, and:

$$\mathbb{P}(z^{(i)} = j) = \phi_j.$$

Once a particular component $z^{(i)} = j$ is chosen, the data point $x^{(i)}$ is drawn from a Gaussian distribution:

$$x^{(i)} | z^{(i)} = j \sim \mathcal{N}(\mu_j, \Sigma_j).$$

Generative Story

The generative process of the model is as follows:

1. Sample a latent variable $z^{(i)} \sim \text{Multinomial}(\phi)$.
2. Given $z^{(i)} = j$, sample $x^{(i)} \sim \mathcal{N}(\mu_j, \Sigma_j)$.

This model is known as the **Mixture of Gaussians**. The latent variables $z^{(i)}$ are not observed in the data, making them *hidden* or *latent* random variables. This unobservability is what makes parameter estimation more complex.

Model Parameters

The full set of parameters to estimate in the model are:

- ϕ : the mixture weights (prior probabilities of each cluster),
- $\mu_1, \dots, \mu_k$: the means of each Gaussian component,
- $\Sigma_1, \dots, \Sigma_k$: the covariance matrices.

Likelihood of the Observed Data

Since the latent variables $z^{(i)}$ are not observed, the likelihood of the observed data involves marginalizing over them:

$$p(x^{(i)}) = \sum_{j=1}^k \phi_j \cdot \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j).$$

The full data likelihood is then:

$$L(\phi, \mu, \Sigma) = \prod_{i=1}^n p(x^{(i)}) = \prod_{i=1}^n \sum_{j=1}^k \phi_j \cdot \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j).$$

Challenge in Estimation

If we had access to the true values of $z^{(i)}$, parameter estimation would be straightforward. However, since $z^{(i)}$ is latent, we need to estimate both:

- the parameters ϕ, μ, Σ ,
- and the most likely hidden assignments $z^{(i)}$.

This challenge is typically addressed using the **Expectation-Maximization (EM)** algorithm.